CRPS & CRPSS — reference (not on poster)

All formulas are for **one gridpoint, one lead week, one initialisation**. Each value (f_j, o, c) is a weekly mean temperature in °C. Subscript j indexes ensemble members ($j = 1, \ldots, M$): there are 11 forecast values but only **one** observed value, so o has no member subscript.

Step 1: Regular CRPS (unfair)

$$\text{CRPS} = \underbrace{\frac{1}{M}\sum_{j=1}^{M}|f_j - o|}_{\text{MAE term}} - \underbrace{\frac{1}{2M^{\mathbf{2}}}\sum_{j=1}^{M}\sum_{k=1}^{M}|f_j - f_k|}_{\text{spread term}}$$

f_j = weekly mean of ensemble member j
o = weekly mean of ERA5 observation
M = ensemble size (11 for EEFH)

The spread term is the mean pairwise distance between all members.

Intuition: how wide is the ensemble?

Problem: with finite M, the double sum underestimates the true spread (you're sampling 11 members from a continuous distribution, so you miss the tails). Smaller ensembles get penalised more.

Step 2: Fair CRPS (Ferro 2014) — what we actually compute

$$\text{CRPS}_{\text{fair}} = \underbrace{\frac{1}{M}\sum_{j=1}^{M}|f_j - o|}_{\text{MAE term (same)}} - \underbrace{\frac{1}{2M(\mathbf{M-1})}\sum_{j=1}^{M}\sum_{k=1}^{M}|f_j - f_k|}_{\text{spread term (corrected)}}$$

Only change: $M^2 \rightarrow M(M-1)$ in the spread denominator.

This inflates the spread credit to compensate for finite ensemble size. Result: 11-member EEFH and 25-member SEAS5 are directly comparable.

Ferro (2014), Eq. 3 [4]

Equivalent sorted form (what the code uses, faster to compute): $\text{CRPS}_{\text{fair}} = \frac{1}{M}\sum_{j=1}^{M}|f_j - o| - \frac{1}{M(M-1)}\sum_{j=1}^{M}(2j - M - 1)\, f_{(j)}$ where $f_{(j)}$ = members sorted ascending.

Step 3: CRPSS — skill relative to climatology

$$\text{CRPSS} = 1 - \frac{\overline{\text{CRPS}}_{\text{fc}}}{\overline{\text{CRPS}}_{\text{clim}}} = 1 - \frac{\frac{1}{N}\sum_{n=1}^{N}\text{CRPS}_{\text{fair}}^{(n)}}{\frac{1}{N}\sum_{n=1}^{N}|c_n - o_n|}$$

$\overline{\text{CRPS}}_{\text{fc}}$ = mean fair CRPS across inits
$\overline{\text{CRPS}}_{\text{clim}}$ = mean $|c - o|$ across inits
c_n = climatological weekly mean (1991–2020)
o_n = observed weekly mean for init n
N = number of initialisations

Why $|c_n - o_n|$ for climatology? Climatology is a single value (M=1). With one "member", the spread term has $M(M-1) = 0$ in the denominator and vanishes. CRPS for a deterministic forecast = absolute error. So the reference is just: how far is the long-term average from the truth?

How to read these

CRPS (lower = better, units °C)

MAE term: average |member − truth| across all 11 members. Large = members far from observation.

Spread term: mean pairwise distance between members, divided by 2. Wider spread → larger subtraction → lower CRPS → rewarded.

The balance: a tight cluster around the wrong value scores badly (large MAE, small spread credit). A wide spread that brackets the truth scores well (moderate MAE, large spread credit).

CRPSS (higher = better, dimensionless)

Numerator (of fraction): mean fair CRPS of the 11-member ensemble, pooled across all inits at one gridpoint.

Denominator (of fraction): mean $|c - o|$ — climatological mean treated as a single deterministic "forecast". No spread term, just absolute error.

CRPSS > 0: ensemble beats climatology.
CRPSS = 0: no better than the long-term average.
CRPSS < 0: climatology wins — forecast is miscalibrated.

Why positive CRPSS where ACC ≈ 0: the ensemble *mean* has no spatial pattern skill, but the 11 members still bracket the truth better than a single climatological value. The spread term is doing the work.